NP-completeness of 3-SAT
COMS20017 (Algorithms and Data)

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An easier problem to reduce from: 3-SAT

A **literal** is either a variable x or its negation $\neg x$. An **OR clause** is an OR of literals. A **CNF** formula is an AND of OR clauses. The **SAT** problem asks: "Is the input CNF formula satisfiable?"

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Certainly 3-SAT $\in$ NP, but our proof that SAT is NP-hard breaks for 3-SAT. So to prove NP-hardness, we will reduce SAT to 3-SAT; the result then follows since SAT is NP-hard.

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C_i has width 2: Say $C_i = x \vee y$. Then we would like to replace C_i with $x \vee y \vee \texttt{False}$ in F', since this is $\texttt{True}$ if and only if $x \vee y = \texttt{True}$.

But $\texttt{False}$ is not a literal... Can we add a new variable which is always $\texttt{False}$ in any satisfying assignment?

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But $\texttt{False}$ is not a literal... Can we add a new variable which is always $\texttt{False}$ in any satisfying assignment? Yes! If we add this CNF to F:

$$F_z = (\neg z_1 \vee z_2 \vee z_3) \wedge (\neg z_1 \vee z_2 \vee \neg z_3) \wedge (\neg z_1 \vee \neg z_2 \vee z_3) \wedge (\neg z_1 \vee \neg z_2 \vee \neg z_3)$$

then z_1 is forced to be $\texttt{False}$: No matter what value z_2 and z_3 take, their literals must both be $\texttt{False}$ in one of the above OR clauses. ✓

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If C_i has width 3: We can just leave it as it is. ✓

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Width 1 clauses: ✓ **Width 2 clauses:** ✓ **Width 3 clauses:** ✓

If C_i has width $k \geq 4$: Say $C_i = \ell_1 \vee \cdots \vee \ell_k$. We would like to replace

$$C_i \to (e_1 = \ell_1 \vee \ell_2) \wedge (e_2 = e_1 \vee \ell_3) \wedge \cdots \wedge (e_{k-2} = e_{k-3} \vee \ell_{k-1}) \wedge (e_{k-2} \vee \ell_k),$$

as given the values of $\ell_1, \ldots, \ell_k$, this is satisfiable if and only if $\ell_1 \vee \cdots \vee \ell_k = \texttt{True}$.

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as given the values of $\ell_1, \ldots, \ell_k$, this is satisfiable if and only if $\ell_1 \vee \cdots \vee \ell_k = \texttt{True}$. How do we implement the e_i's? We have

$$(a = b \vee c) \text{ if and only if } (a \vee \neg b) \wedge (a \vee \neg c) \wedge (\neg a \vee b \vee c);$$

the first two clauses on the right enforce $a = \texttt{False} \Rightarrow b \vee c = \texttt{False}$, and the last enforces $b \vee c = \texttt{False} \Rightarrow a = \texttt{False}$. □

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

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First we note what we'd like the clauses to become.

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$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = {} & (u \vee f_1 \vee \texttt{False}) \wedge (\neg u \vee \neg v \vee f_1) \\ & \wedge (e_1 = v \vee \neg w) \wedge (e_2 = e_1 \vee x) \wedge (e_3 = e_2 \vee \neg y) \\ & \wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ & \wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ & \wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge \boldsymbol{(\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2)} \\ & \wedge \boldsymbol{(\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2)}. \end{aligned}$$

We simulate the first instance of `False` with f_1... and the second instance with f_2.

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = (u \vee f_1 \vee \boldsymbol{f_2}) &\wedge (\neg u \vee \neg v \vee f_1) \\ &\wedge (e_1 = v \vee \neg w) \wedge (e_2 = e_1 \vee x) \wedge (e_3 = e_2 \vee \neg y) \\ &\wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ &\wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ &\wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ &\wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

We simulate the first instance of `False` with f_1... and the second instance with f_2.

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = {} & (u \vee f_1 \vee f_2) \wedge (\neg u \vee \neg v \vee f_1) \\ & \wedge (e_1 = v \vee \neg w) \wedge (e_2 = e_1 \vee x) \wedge (e_3 = e_2 \vee \neg y) \\ & \wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ & \wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ & \wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ & \wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

Finally, we simulate the equality clauses.

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = (u \vee f_1 \vee f_2) &\wedge (\neg u \vee \neg v \vee f_1) \\ &\wedge (e_1 = v \vee \neg w) \\ &\wedge (e_2 = e_1 \vee x) \\ &\wedge (e_3 = e_2 \vee \neg y) \\ &\wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ &\wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ &\wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ &\wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

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Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = {} & (u \vee f_1 \vee f_2) \wedge (\neg u \vee \neg v \vee f_1) \\ & \wedge \boldsymbol{(e_1 \vee \neg v \vee f_1) \wedge (e_1 \vee w \vee f_1) \wedge (\neg e_1 \vee v \vee \neg w)} \\ & \wedge (e_2 = e_1 \vee x) \\ & \wedge (e_3 = e_2 \vee \neg y) \\ & \wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ & \wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ & \wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ & \wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

Finally, we simulate the equality clauses.

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = {} & (u \vee f_1 \vee f_2) \wedge (\neg u \vee \neg v \vee f_1) \\ & \wedge (e_1 \vee \neg v \vee f_1) \wedge (e_1 \vee w \vee f_1) \wedge (\neg e_1 \vee v \vee \neg w) \\ & \wedge \boldsymbol{(e_2 \vee \neg e_1 \vee f_1) \wedge (e_2 \vee \neg x \vee f_1) \wedge (\neg e_2 \vee e_1 \vee x)} \\ & \wedge (e_3 = e_2 \vee \neg y) \\ & \wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ & \wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ & \wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ & \wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

Finally, we simulate the equality clauses.

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = {} & (u \vee f_1 \vee f_2) \wedge (\neg u \vee \neg v \vee f_1) \\ & \wedge (e_1 \vee \neg v \vee f_1) \wedge (e_1 \vee w \vee f_1) \wedge (\neg e_1 \vee v \vee \neg w) \\ & \wedge (e_2 \vee \neg e_1 \vee f_1) \wedge (e_2 \vee \neg x \vee f_1) \wedge (\neg e_2 \vee e_1 \vee x) \\ & \wedge \boldsymbol{(e_3 \vee \neg e_2 \vee f_1) \wedge (e_3 \vee y \vee f_1) \vee (\neg e_3 \vee e_2 \vee \neg y)} \\ & \wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ & \wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ & \wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ & \wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

Finally, we simulate the equality clauses.

Example of SAT $\leq_c$ 3-SAT reduction

Suppose our original SAT instance is:

$$F = u \wedge (\neg u \vee \neg v) \wedge (v \vee \neg w \vee x \vee \neg y \vee \neg z) \wedge (y \vee z) \wedge (\neg v \vee w \vee z).$$

We transform this into a 3-SAT instance as follows:

$$\begin{aligned} F' = (u \vee f_1 \vee f_2) &\wedge (\neg u \vee \neg v \vee f_1) \\ &\wedge (e_1 \vee \neg v \vee f_1) \wedge (e_1 \vee w \vee f_1) \wedge (\neg e_1 \vee v \vee \neg w) \\ &\wedge (e_2 \vee \neg e_1 \vee f_1) \wedge (e_2 \vee \neg x \vee f_1) \wedge (\neg e_2 \vee e_1 \vee x) \\ &\wedge (e_3 \vee \neg e_2 \vee f_1) \wedge (e_3 \vee y \vee f_1) \vee (\neg e_3 \vee e_2 \vee \neg y) \\ &\wedge (e_3 \vee \neg z \vee f_1) \wedge (y \vee z \vee f_1) \wedge (\neg v \vee w \vee z) \\ &\wedge (\neg f_1 \vee a_1 \vee a_2) \wedge (\neg f_1 \vee a_1 \vee \neg a_2) \wedge (\neg f_1 \vee \neg a_1 \vee a_2) \\ &\wedge (\neg f_1 \vee \neg a_1 \vee \neg a_2) \wedge (\neg f_2 \vee a_1 \vee a_2) \wedge (\neg f_2 \vee a_1 \vee \neg a_2) \\ &\wedge (\neg f_2 \vee \neg a_1 \vee a_2) \wedge (\neg f_2 \vee \neg a_1 \vee \neg a_2). \end{aligned}$$

Phew! We could have done this directly, without the gadgets as intermediate steps, but they made it much easier to think about...